

(idiosyncratic) risk from the order and will more often trade in an aggressive manner and incur higher cost. Next suppose that the trade list consists of the same 500,000 share buy order for RLK and an additional 100 buy orders. Here the manager achieves diversification of company specific risk due to the large number of names in the trade basket and hence is only exposed primarily to market risk. The manager could trade this list at a more moderate rate since there is less risk exposure (only one source of risk as opposed to two sources of risk). This results in a lower market impact cost for the 500,000 shares of RLK. Finally, suppose the manager performs a rebalance of the portfolio and the trade list is comprised of the 500,000 buy order for RLK plus an additional 100 stocks to buy and an additional 100 stocks to sell (with equal dollars across both the buy list and sell list). This trade list will now achieve risk reduction from diversification of company specific risk just like it did in the previous scenario and will also achieve market risk reduction from having a two-sided portfolio (due to the buy and sell orders). Now RLK can be traded in a passive manner and the corresponding trading cost will be even lower.

The second wave of portfolio optimizers did not take into account the corresponding trading cost resulting from the actual implementation strategy—which as we demonstrated above can vary dramatically. The second wave of optimizers would assess the same exact cost to the order regardless of the other names in the trade list and regardless of how those shares would be transacted. Aggressive, moderate, and passive strategies would all be assumed to incur the same trading cost. Finally, the second wave of portfolio optimizers did not provide managers or traders with any insight at all into how the targeted portfolio should be best implemented. It is left to the managers and traders to determine. But as we have discussed above, the goals of these parties are often conflicting.

Third Wave: The third wave of portfolio optimization with transaction costs consists of incorporating a market impact function that is dependent upon the size of the order, the overall risk composition of the trade list, and the underlying trade schedule. This portfolio optimization problem has been studied by Engle and Ferstenberg (2006, 2007) and Kissell and Malamut (2006). The advantages of this type of optimization are (1) it will properly account for the trading cost based on the underlying trading strategy, and (2) it will provide as output from the process the exact trading schedule to achieve the targeted portfolio so there will be perfect alignment between portfolio manager and trader. For example, these optimization processes will have different costs for the 500,000 share order